

TITLE: Extracorporeal Membrane Oxygenation (ECMO) in Trauma

PROTOCOL: This protocol serves to outline inclusion and exclusion criteria for the use of ECMO in trauma patients. It also describes team dynamics and logistical strategy to ensure seamless transition and continued support among clinicians in the trauma resuscitation bay, trauma intensive care unit, unit 87, and ERS team.

PURPOSE: The use of Extracorporeal Membrane Oxygenation (ECMO) in the setting of trauma is supported by evidence of its safety and effectiveness. The number of Extracorporeal Life Support Organization (ELSO) ECMO centers and reported ECMO runs has increased over the last decade. The number of trauma-ECMO runs reported by TQIP data, however, has remained stable over the last 5 years. This represents the potential to expand its use in the trauma patient population at UF Health Shands.

APPROVED:



Liza Laquain, MD
Co-Chair ICU Protocol Committee



Lindsey Kranson, PA-C
Co-Chair ICU Protocol Committee

DEFINITIONS:

- A. ECMO - Extracorporeal Membrane Oxygenation
- B. V-V ECMO – Veno-venous Extracorporeal Membrane Oxygenation
- C. V-A ECMO – Veno-arterial Extracorporeal Membrane Oxygenation
- D. ERS – Emergency Resuscitation Service
- E. ISS – Injury Severity Score
- F. TRALI – Transfusion Associated Acute Lung Injury
- G. AIS – Abbreviated Injury Scale
- H. TICU – Trauma Intensive Care Unit
- I. CCM – Critical Care Medicine
- J. TQUIP – Trauma Quality Improvement Program

CORE PROCEDURE:

I. Candidacy

- A. Inclusion Criteria
 - a. Age 16 to 70
 - b. If active bleeding, hemostasis achievable pending operative or interventional management.
 - c. V-V ECMO-Specific Criteria
 - i. Refractory respiratory failure with maximal lung protective support
 - ii. P/F Ratio <100
 - iii. Respiratory acidosis (pH <7.20)
 - iv. Elevated plateau pressures (≥ 30 cm H₂O)
 - v. Lung lacerations or bronchopulmonary fistulas in patients requiring positive pressure ventilation
 - vi. Temporary support to allow for airway injury exploration and management
 - vii. Bridge to Recovery (e.g. pulmonary embolism, pulmonary contusion, TRALI)
 - viii. Bridge to Transplant
 - d. V-A ECMO-Specific Criteria
 - i. Cardiogenic shock secondary to a reversible etiology (e.g. stress-induced cardiomyopathy, pulmonary embolism)
 - ii. Non-recrutable myocardium after inotrope challenge
 - iii. Bridge to recovery, transplant, or durable mechanical device

B. Relative Exclusion Criteria

*Consults outside of these relative exclusion criteria will be considered on a case-by-case basis considering mode of trauma, Injury Severity Score (ISS), Thoracic Abbreviated Injury Scale (AIS), survival score (Murray, Respiratory ECMO Survival Prediction, Survival after Veno-Arterial ECMO), days on ventilator, etiology of respiratory failure/shock, etc.

- a. Severe respiratory failure ≥ 7 days
- b. Non-reversible etiology of respiratory failure
- c. High pre-injury morbidity
- d. High physiologic frailty
- e. Terminal disease
- f. Metastatic malignancy
- g. Devastating neurologic injury
- h. Advanced multisystem organ failure

II. Consultation Initiation

- A. Consultations to the Emergency Resuscitation Service (ERS) may be initiated by, or at the request of the attending trauma surgeon or intensivist.
- B. The primary service or TICU CCM team member will call the ERS First On-Call physician at (352) 283-4198.
- C. Consultations outside of the outlined inclusion/exclusion criteria may be initiated at the discretion of the primary service or critical care medicine service and will be considered on a case-by-case basis.
- D. The decision to initiate ECMO will be made by the ERS Senior On-Call physician.
- E. At the time of ECMO consideration, no further attempts to obtain vascular access in either groin shall be made to preserve vascular anatomy.

III. Cannulation of the ECMO Candidate

- A. Upon determination of candidacy, the ERS attending will summon the ECMO cannulation team including the ECMO Primer.
- B. The ERS attending will call the Charge RN on unit 87 to facilitate obtaining a bed space.
- C. ERS attending will determine the resources needed and appropriate location of cannulation (e.g. bedside, resuscitation bay, fluoroscopy, TEE).
- D. ERS will consult thoracic surgery team for co-management.
- E. ERS on call will place initial orders including labs and imaging as per their protocol

IV. ICU Transfer

- A. Post-cannulation, ERS will become the primary team.
- B. When deemed stable for transport by ERS attending, ECMO patients will be transferred to unit 87.
- C. The trauma service will continue to follow the patient for injury management recommendations.
- D. For all urgent and emergent changes in patient status, the 87 CCM team will notify the Trauma Attending directly at 352-682-1027.

V. Anticoagulation

- A. Anticoagulation decisions will be carefully evaluated on a case-by-case basis, balancing the risk of hemorrhage against the risk of circuit and systemic thrombosis.
- B. The trauma service will determine whether the risk of hemorrhage precludes anticoagulation, which the ERS attending will consider when assessing ECMO candidacy.
- C. Once a joint decision is made by ERS and trauma, the ERS service will oversee the day-to-day management of anticoagulation in consultation with the trauma service.

VII. Transfer of Care

- A. Once the patient is weaned from ECMO support and decannulated from ECMO, the patient will be signed out to TICU by ERS service and transferred to TICU. Care will be transitioned to trauma service and ERS will sign off as appropriate.

Protocol Summary

Inclusion Criteria	Relative Exclusion Criteria*
- Refractory respiratory failure with maximal lung protective support and / or - Cardiogenic shock secondary to reversible etiology (e.g. stress-induced cardiomyopathy, pulmonary embolism)	- Severe respiratory failure ≥ 7 days - Non-reversible etiology of respiratory failure
Age 16 to 70	High pre-injury morbidity, high physiologic frailty, terminal disease, metastatic malignancy
If active bleeding, hemostasis achievable (pending OR or IR)	Refractory hemorrhagic shock
	Devastating neurologic injury
	Advanced multisystem organ failure

*Consults outside of these relative exclusion criteria will be considered on a case-by-case basis considering mode of trauma, Injury Severity Score (ISS), Thoracic Abbreviated Injury Scale (AIS), survival score (Murray, Respiratory ECMO Survival Prediction, Survival after Veno-Arterial ECMO), days on ventilator, etiology of respiratory failure/shock, etc.

REFERENCES:

- Biderman, P., et al. Extracorporeal life support in patients with multiple injuries and severe respiratory failure: a single-center experience? The Journal of Trauma and Acute Care Surgery, 2013. 75(5): p. 907–912.
- Michaels, A.J., et al. Extracorporeal life support in pulmonary failure after trauma. The Journal of Trauma, 1999. 46(4): p. 638– 645.

Schmidt M, et al. Predicting survival after extracorporeal membrane oxygenation for severe acute respiratory failure. The Respiratory Extracorporeal Membrane Oxygenation Survival Prediction (RESP) score. *Am J Respir Crit Care Med*. 2014 Jun 1;189(11):1374-82. doi: 10.1164/rccm.201311-2023OC. PMID: 24693864.

Peetermans M, et al. Impact of BMI on outcomes in respiratory ECMO: an ELSO registry study. *Intensive Care Med*. 2023 Jan;49(1):37-49. doi: 10.1007/s00134-022-06926-4. Epub 2022 Nov 22. PMID: 36416896; PMCID: PMC9684759.

Muellenbach RM, et al. Prolonged heparin-free extracorporeal membrane oxygenation in multiple injured acute respiratory distress syndrome patients with traumatic brain injury. *J Trauma Acute Care Surg*. 2012 May;72(5):1444-7. doi: 10.1097/TA.0b013e31824d68e3. PMID: 22673280.

Parker BM, Menaker J, Berry CD, et al. Single Center Experience With Veno-Venous Extracorporeal Membrane Oxygenation in Patients With Traumatic Brain Injury. *Am Surg*. 2021;87(6):949-953. doi:10.1177/0003134820956360

ECMO Registry Report, Extracorporeal Life Support Organization. [elseo.org](https://www.elseo.org)